Module: 2 Algebraic Structures

Introduction:

- **Uniary Operations:** Let S be any set. we say that * is an uniary operation on S if $*: S \to S$

- **Binary Operation:** If $*: S \times S \to S$, then $*$ is called as binary operation. Also we can say that $*$ satisfies the closure property.

- **Algebraic Systems:** A system consisting of a set and one or more operations on the set S is called an algebraic system or simply algebra.

- **Notation:** $\langle S, f_1, f_1, \ldots, f_n \rangle$, where S is a nonempty set and $f_1, f_2, \ldots f_n$ are operations on S.

- **Algebraic structures:** Since the operations and relations on the set S define a structure on the elements of S, then an algebraic system is called an algebraic structure.

Example: 1 Let I be the set of Integers.

Consider the algebraic system $\langle I, +, \times \rangle$ where $+$ and $\times$ are the operations of addition and multiplication on I.

A list of important properties of $\langle I, +, \times \rangle$

(A1) Associativity

For any $a, b, c \in I$

$$(a+b)+c = a+(b+c)$$

(A2) Commutativity

For any $a, b \in I$, then $a+b = b+a$.

(A3) Identity element

$\exists$ a distinguished element $0 \in I$ s.t for any $a \in I$

$$a+0 = 0+a = a$$

Here $0 \in I$ is the identity element w.r.to addition.

(A4) Inverse element

For each $a \in I$, $\exists$ a element different from identity element in I denoted by '$-a$' and called the negative of a (or) additive inverse of a s.t $a+(-a) = 0$.

(M1) Associativity

For $a, b, c \in I$, $(a \times b) \times c = a \times (b \times c)$

(M2) Commutativity

For $a, b \in I$, $a \times b = b \times a$

(M3) Identity element

There exists a distinguished element $1 \in I$ s.t for any $a \in I$,

$$a \times 1 = 1 \times a = a$$

(D) Distributivity

For any $a, b, c \in I$,

$$a \times (b+c) = (a \times b) + (a \times c)$$

The operation $\times$ distributes over $+$.

(C) Cancellation property

For any $a, b, c \in I$ and $a \neq 0$.

$$a \times b = a \times c \Rightarrow b = c.$$

Example 2.

Let $\mathbb{R}$ be the set of real numbers and $+$ and $\times$ be the operations of addition and multiplication on $\mathbb{R}$. The algebraic systems $\langle \mathbb{R}, +, \times \rangle$ satisfies all the properties given for the systems $\langle I, +, \times \rangle$.

Semigroups and Monoids

Semigroup : Let S be a nonempty set and $\circ$ be a binary operation on S. The algebraic system $\langle S, \circ \rangle$ is called a semigroup if the operation $\circ$ is associative.

In other words, $\langle S, \circ \rangle$ is a semigroup if for any $x, y, z \in S$, $(x \circ y) \circ z = x \circ (y \circ z)$.

Monoid A Semigroup $\langle M, \circ \rangle$ with an identity element with respect to the operation $\circ$ is called a monoid. In other words, an algebraic system $\langle M, \circ \rangle$ is called a monoid if for any $x, y, z \in M$,

$$(x \circ y) \circ z = x \circ (y \circ z)$$

and $\exists$ an elt $e \in M$ s.t for any $x \in M$,

$$e \circ x = x \circ e = x.$$

Note: (1) An identity elt of any binary operation, if it exists, is unique.

(2) Sometimes represent a monoid as $\langle M, \circ, e \rangle$ to emphasize the fact that e is a distinguished elt of such a monoid.

Example: (1)

Let X be a nonempty set and X^X be the set of all mappings from X to X.

Let $\circ$ denote the operation of composition of the mappings, ie, for $f, g \in X^X$, $f \circ g$ is given by

$$f \circ g (x) = f(g(x)) \ \forall \ x \in X \text{ is in } X^X.$$

$$\text{ie, } f \circ g \in X^X$$

$\therefore$ the algebra $\langle X^X, \circ \rangle$ is a monoid, because the operation of composition is associative and the identity mapping $f(x) = x \ \forall \ x \in X$ is the identity of the operation.

(2) Let X be a nonempty set and $B(X)$ be the set of all relations from X to X.

Let '$\circ$' denote the composition operation of relations in $B(X)$ then $\langle B(X), \circ \rangle$ is a monoid in which the identity relation is the identity of the monoid.

(3) Let S be a nonempty set and $P(S)$ be its power set. The algebras $\langle P(S), \cup \rangle$ & $\langle P(S), \cap \rangle$ are monoids with the identities ϕ & S respectively.

④ * Let N be the set of all natural nos and '0'. Then $\langle N, + \rangle$ and $\langle N, \times \rangle$ are monoids with the identities 0 and 1 respectively.

* Let E denotes the set of all positive even numbers. Then $\langle E, + \rangle$ & $\langle E, \times \rangle$ are semigroups but not monoids.

Homomorphisms of Semigroups and Monoids

Semigroup Homomorphism

Let $\langle S, * \rangle$ and $\langle T, \Delta \rangle$ be any two semigroups. A mapping $g : S \to T$ is called semigroup homo., if for any $a, b \in S$,

$$g(a * b) = g(a) \Delta g(b).$$

Note:- A semigroup homo., is called a

(i) Semigroup monomorphism, if the mapping is 'one to one',

(ii) Semigroup epimorphism, if the mapping is 'onto'

and (iii) Semigroup isomorphism, if the mapping is 1-1 & onto.

Isomorphic

Two semigroups $\langle S, * \rangle$ and $\langle T, \Delta \rangle$ are said to be isomorphic, if there exists a semigroup isomorphism b/w S and T.

Note: (1) Homomorphism preserves the semigroup character because it preserves associativity

(2) Semigroup preserves the property of idempotency & commutativity.

Definition Monoid Homomorphism

Let $\langle M, *, e_M\rangle$ and $\langle T, \Delta, e_T\rangle$ be any two monoids. A mapping $g : M \to T$ s.t for any two elts $a, b \in M$ s.t

$$g(a * b) = g(a)\, \Delta\, g(b) \quad \text{and}$$

$$g(e_M) = e_T$$ is called a monoid homo.,

Note: The monoid homomorphism preserves

(1) associativity

(2) Commutativity

(3) identity

(4) Invertibility.

Example: ①

Let $\langle \mathbb{N}, +\rangle$ be the semigroup of natural numbers with '0', and $\langle S, *\rangle$ be the semigroup on $S = \{e, 0, 1\}$ with the operation $*$ given by following table

$g(a+b) = g(a) * g(b)$

*	e	0	1
e	e	0	1
0	0	0	0
1	1	0	1

$g(0+1) = g(0) * g(1)$
$1 * 0 = 0 \in S$

A mapping $g : \mathbb{N} \to S$ given by $g(0) = 1$ & $g(j) = 0$ for $j \neq 0$ is a semigp homomorphism.

Both $\langle N, +\rangle$ and $\langle S, *\rangle$ are monoids with identities 0 and e respectively, but g is not a monoid homo., because $g(0) \neq e$.

Theorem : 1 Let $\langle S, *\rangle$, $\langle T, \Delta\rangle$ and $\langle V, \oplus\rangle$ be semigroups and $g: S \to T$, $h: T \to V$ be semigroup homomorphisms. Then $(h \circ g): S \to V$ is a semi gp homo., from $\langle S, *\rangle$ to $\langle V, \oplus\rangle$.

(or) composition of two homo is also a homo.

Soln:- Let $a, b \in S$. Then

$$\begin{aligned}(h \circ g)(a * b) &= h\,g(a*b)\\ &= h(g(a) \Delta g(b))\\ &= h\,g(a) \oplus h\,g(b)\\ &= (h \circ g)(a) \oplus (h \circ g)(b).\ //\end{aligned}$$

Definition:- A homomorphism of a semigroup into itself is called a semi gp endomorphism, while a isomorphism onto itself is called a semigroup automorphism.

Theorem : 2 Let $\langle S, *\rangle$ be a given semigroup. There exists a homomorphism $g: S \to S^S$ where $\langle S^S, \circ\rangle$ is a semi gp of functions from S to S under the operation of (left) composition.

Proof:- Let $a \in S$ be a fixed elt.

Let $g(a) = f_a$ where $f_a \in S^S$ & f_a is defined by $f_a(b) = a * b$ for any $b \in S$.

Also $g(a*b) = f_{a*b}$

Now
$$\begin{aligned}f_{a*b}(c) &= (a*b)*c\\ &= a*(b*c)\\ &= a * f_b(c)\\ &= f_a(f_b(c))\\ &= (f_a \circ f_b)(c)\ \forall\, c \in S.\end{aligned}$$

Therefore $g(a*b) = f_{a*b}$

$= f_a \circ f_b$

$= g(a) \circ g(b)$.

$\therefore$ $g: S \to S^S$ is a homomorphism of $(S, *)$ into $\langle S^S, \circ \rangle$.

~~Theorem~~ Sub Semigroups and Submonoids

Sub semigroups: Let $\langle S, * \rangle$ be a Semigroup. and $T \subseteq S$. If the set T is closed under the operation, $*$, then $\langle T, * \rangle$ is said to be Sub Semigroup of $\langle S, * \rangle$.

i.e, If $T \subseteq S$ then $\forall\, a, b \in T \Rightarrow a*b \in T$.

Submonoid: Let $\langle M, *, e \rangle$ be a monoid and $T \subseteq M$. If T is closed under the operations $*$ and $e \in T$, then $\langle T, *, e \rangle$ is said to be submonoid of $\langle M, *, e \rangle$.

i.e, If $T \subseteq M$ then $\forall\, a, b \in T \Rightarrow a*b \in T$ & $e \in T$.

Example: 1 For ~~any~~ the semigroup $\langle N, \times \rangle$, let T be the set of multiples of the integer m.

Then clearly $T \subseteq N$, Also $\langle T, \times \rangle$ is a subsemigp of $\langle N, \times \rangle$.

For, $a, b \in T \Rightarrow a = k_a m$, $b = k_b \cdot m$ where $k_a, k_b \in N$

Then $a \times b = k_a m \times k_b \cdot m$

$= (k_a k_b\, m) m \in T$

$\therefore$ $\langle T, \times \rangle$ is a subsemigroup of $\langle N, \times \rangle$.

(2) For the semigroup $\langle N, + \rangle$, the set E of all the even non-negative integers is a subsemigroup $\langle E, + \rangle$ of $\langle N, + \rangle$.

Proof:- Let $a, b \in T \Rightarrow a = 2m$ & $b = 2n$ where $m, n \geq 0$ & $m, n \in \mathbb{Z}$

Then $a+b = 2m+2n = 2(m+n) \in E$

Hence $\langle E, + \rangle$ is a subsemigp of $\langle N, + \rangle$.

(3) Let $\langle S, * \rangle$ be a semigp and monoid on $S = \{e, 0, 1\}$ with the identity 'e', defined by

*	e	0	1
e	e	0	1
0	0	0	0
1	1	0	1

Take $S' = \{0, 1\} \subseteq S = \{e, 0, 1\}$

Then $\langle S', * \rangle$ is a subsemigp of $\langle S, * \rangle$ but not a submonoid of $\langle S, * \rangle$.

Proof:- Here $0 * 1 = 0 \in S' \Rightarrow \langle S', * \rangle$ is a subsemigp of $\langle S, * \rangle$

But $e \notin S' \Rightarrow \langle S', * \rangle$ is not a submonoid of $\langle S, * \rangle$.

Commutative semigroup: A semigp $\langle S, * \rangle$ is said to be commutative semigp if for any $a, b \in S$ s.t $a * b = b * a$.

Eg: $\langle N, + \rangle$

Commutative Monoid: A monoid $\langle M, *, e \rangle$ is said to be commutative monoid if for any $a, b \in M$ s.t

$a * b = b * a$ Eg: $\langle N, + \rangle$

Idempotent elt

Suppose $\langle S, * \rangle$ is a algebraic system and $a \in S$.

If $a * a = a$ then $a \in S$ is said to be idempotent elt.

Eg: Identity elt.

Theorem For any commutative monoid $\langle M, * \rangle$, the set of idempotent elements of M forms a submonoid.

Proof:- Let S be the set of idempotent elts of M.

To prove $\langle S, * \rangle$ is a submonoid of $\langle M, * \rangle$

Since, the identity elt 'e' of M is idempotent

then $e \in S$. $\because (e * e = e)$

It is enough to prove 'S' is closed under $*$.

let $a, b \in S \Rightarrow a * a = a$ & $b * b = b$.

$$
\begin{aligned}
(a * b) * (a * b) &= (a * b) * (b * a) \\
&= a * (b * b) * a \\
&= a * b * a \\
&= a * \cancel{a} * b \\
&= a * b
\end{aligned}
$$

$\therefore$ $a * b \in S$.

Hence $\langle S, * \rangle$ is a submonoid of $\langle M, *, e \rangle$.

Direct product of Algebraic systems

Let $\langle S, * \rangle$ and $\langle T, \Delta \rangle$ be two algebraic systems. The direct product of $\langle S, * \rangle$ & $\langle T, \Delta \rangle$ is the algebraic system $\langle S \times T, \circ \rangle$ in which the operation '$\circ$' on $S \times T$ is defined by

$$\langle s_1, t_1 \rangle \circ \langle s_2, t_2 \rangle = \langle s_1 * s_2, t_1 \Delta t_2 \rangle$$

for any $\langle s_1, t_1 \rangle$ & $\langle s_2, t_2 \rangle \in S \times T$.

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Groups

Definition and examples

A Group $\langle G, * \rangle$ is an algebraic system in which the binary operation $*$ on G Satisfies three conditions

(1) Associativity

For all $x, y, z \in G$,

$$x * (y * z) = (x * y) * z$$

(2) Identity

$\exists$ an elt $e \in G$ s.t for any $x \in G$,

$$x * e = e * x = x$$

(3) Inverse

$\forall\ x \in G$, $\exists$ an elt $x^{-1} \in G$ s.t

$$x^{-1} * x = x * x^{-1} = e.$$

Note:

1) A group $(G, *)$ in which the operation $*$ is commutative is called abelian group (or) Commutative gp.

2) Cancellation property is hold in every gp. (Since the existence of the inverse elt of every elt)

$$\left.\begin{array}{l} a * b = a * c \Rightarrow b = c \\ b * a = c * a \Rightarrow b = c \end{array}\right\} \forall\ a, b, c \in G$$

3) S.T a gp cannot have any elt which is idempotent except the identity.

Proof:- Let $a \in G$ be an idempotent elt of G

i.e., $a * a = a$

Now $e = a * a^{-1} = a^{-1} * a$

$= a^{-1} * (a * a)$

$= (a^{-1} * a) * a$

$= e * a$

$e = a$

Hence the identity is the only idempotent elt of G.

Permutation

Any one-to-one mapping of a set S onto S is called a permutation of S.

Order of a Group $|G|$

The order of a group $\langle G, * \rangle$ denoted by $|G|$, is the number of elts of G which G is finite.

Examples

① Let $\mathbb{I}$ (or $\mathbb{Z}$) be set of integers. $\mathbb{Z} = \{0, \pm 1, \pm 2, \ldots\}$

The algebraic system $\langle \mathbb{Z}, + \rangle$ is an abelian gp.

② $\langle \mathbb{R}, + \rangle$ and $\langle \mathbb{R} - \{0\}, \times \rangle$ are abelian groups where $\mathbb{R}$ is the set of all real numbers

③ $\langle \mathbb{Q}, + \rangle$ and $\langle \mathbb{Q} - \{0\}, \times \rangle$ are abelian groups where $\mathbb{Q}$ is the set of all rational numbers.

④ $\langle \mathbb{Z}_m, \oplus \rangle$ (or) $\langle \mathbb{Z}_m, +_m \rangle$ and

(Z_p\{0}, x_p) are abelian group, p-prime number

where $\mathbb{Z}_m = \{[0], [1], \ldots, [m]\}$

$\oplus$ or $+_m$ = addition modulo 'm'

$\otimes$ or $\times_m$ = multiplication modulo 'm'.

Examples of Groups

1. If M_2 is the set of 2×2 non-singular matrices over $\mathbb{R}$ via $M_2 = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \middle| \, a,b,c,d \in \mathbb{R} \text{ \& } ad-bc \neq 0 \right\}$.

P.T M_2 is a gp under the operation of usual ~~addition~~ matrix multiplication.

Soln:-

If $A = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}$. $B = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}$

clearly $AB = \begin{bmatrix} a_1a_2 + b_1c_2 & a_1b_2 + b_1d_2 \\ c_1a_2 + d_1c_2 & c_1b_2 + d_1d_2 \end{bmatrix} \in M_2$

closure is true.

$\parallel^{ly}$ we can prove Associative

Now $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is the identity elt $\in M_2$

$A^{-1} = \frac{1}{|A|} \operatorname{adj} A$ exists $\in M_2$

$\therefore \langle M_2, \times \rangle$ is a gp.

But $AB \neq BA$ So is not commutative (or) abelian.

2. If $*$ is defined on Q^+ s.t $a*b = \frac{ab}{3}$ for $a,b \in Q^+$, Show that $\{Q^+, *\}$ is an abelian gp.

Soln:-

(i) $a,b \in Q^+$, $\frac{ab}{3} \in Q^+$ closure.

(2) $(a*b)*c = \frac{ab}{3} * c = \frac{ab}{3} \cdot \frac{c}{3} = \frac{abc}{9}$

$a*(b*c) = a * \frac{bc}{3} = \frac{a}{3} \cdot \frac{bc}{3} = \frac{abc}{9}$

$\therefore$ associative.

(3) let e be the identity elt of Q^+ under $*$

s.t $a * e = e * a = a$

$$\frac{ae}{3} = a \qquad e = 3 \in Q^+$$

$ae = 3a$ (or)

$(ae - 3a) = 0$

$\Rightarrow e = 3$

$\therefore$ Identity elt exists

(4) let b be the inverse elt s.t

$a * b = b * a = e = 3$

$$\frac{ab}{3} = 3$$

$b = \frac{9}{a} \in Q^+$ is the inverse of a.

$$\boxed{\therefore a * \frac{9}{a} = \frac{\cancel{a} \times 9 \cancel{3}}{\cancel{3}\cancel{a}} = e}$$

H.W

(3) $a * b = \frac{ab}{2} \quad a, b \in Q^+$.

(4) $a * b = a + b + 1, \; a, b \in \mathbb{Z}$ ✓ $(a * b) * c = (a+b+1) * c$

$= a+b+1+c+1$

$= a+b+c+2$

(5) $a * b = a + b - ab, \; a, b \in \mathbb{R}$

✓ $a * e = a$

$\cancel{a} + e + 1 = \cancel{a}$

$\boxed{e = -1}$

(6) Verify addition modulo and multiplication modulo

check $\langle \mathbb{Z}_6, \times_6 \rangle$ is a group?

$\mathbb{Z}_6 = \{ [0], [1], [2], [3], [4], [5] \}$

✓ $[0] \times_6 [1] = [0]$ closure

✓ $([0] \times_6 [1]) \times_6 [3] = [0] = [0] \times_6 ([1] \times_6 [3])$

✓ $[1]$ identity

✓ a^{-1} be the inverse then $[a] \times_6 [a^{-1}] = [1]$ is not a gp

$[a^{-1}] = \frac{[1]}{[a]}$

but monoid.

Examples Groups and order of Groups

1) $\langle \{e\}, * \rangle \to$ Group of order 1

2) $\langle \{e, a\}, * \rangle \to$ Group of order 2

Where a be any elt s.t $a * e = e * a = a$

3) $\langle \{e, a, b\}, * \rangle \to$ Group of order 3

where $*$ is defined by

*	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

4) $\langle \mathbb{Z}_m, +_m \rangle$

are group of order m.

Permutation groups

The set of all permutations of the elements of a finite set together with a binary operation (For example, right composition) form a group. Such groups are called Permutation groups.

Cyclic groups

A group $\langle G, * \rangle$ is said to be cyclic, If there exists an element $a \in G$ s.t every elt of G can be written as some power of a, that is a^n for some integer 'n'.

Notion:- Let $\langle G, * \rangle$ be a group and $a \in G$.

Define $a^0 = e$, $a^{n+1} = a^n * a$ for $n \in Z$

i.e, $a^n = \underbrace{a * a * \dots * a}_{n\text{-times}}$

also define $a^{-n} = (a^{-1})^n$ for $n \in N$ where a^{-1} is the inverse elt of $a \in G$.

Result ① A cyclic group is abelian

Remark: A cyclic gp is said to be a gp generated by 'a', or 'a' is a generator of the gp G, if $\exists\, a \in G$ s.t every elt of G can be expressed as a^n for some integer $n \in \mathbb{Z}$.

Proof:- Let $\langle G, * \rangle$ be a cyclic group.

Let $a \in G$ be a generator of $\langle G, * \rangle$

Let $b, c \in G$. **To prove:** $b * c = c * a$

Given $b = a^m$, $c = a^n$ for some $m, n \in \mathbb{Z}$

Now

$$b * c = a^m * a^n$$

$$= \underbrace{a * a * \dots * a}_{m \text{ times}} * \underbrace{a * a * \dots * a}_{n \text{ times}}$$

$$= a^{m+n} = a^{n+m}$$

$$= \underbrace{a * a * \dots * a}_{n \text{ times}} * \underbrace{a * a \dots * a}_{m \text{ times}}$$

$$= a^n * a^m$$

$$= c * b$$

Hence $\langle G, * \rangle$ is a commutative (or) abelian group.

Result ② If 'a' is a generator of a cyclic group $\langle G, * \rangle$, then a^{-1} is also a generator of $\langle G, * \rangle$.

Proof:- Let $b \in G$. Then $b = a^m$ for some $m \in \mathbb{Z}$.

Now, $b = a^{-(-m)} = (a^{-1})^{(-m)}$ Since $(a^{-1})^n = a^{-n}$

$= (a^{-1})^{(-m)}$ for some $-m \in \mathbb{Z}$

Hence (a^{-1}) is also a generator for $\langle G, * \rangle$.

Examples for cyclic Groups

(1) Let $\langle G, \times \rangle$ where $G = \{1, -1, i, -i\}$ is a gp with usual multiplication $(\times)$. Now $i^0 = 1 \in G$, $i^1 = i$, $i^2 = -1$, $i^3 = -i \in G$

Hence $\langle G, \times \rangle$ is a cyclic gp with the generator 'i'.

(2) $\langle \mathbb{Z}_6, +_6 \rangle$ is a cyclic gp with [1] & [5] are generators.

Order of the element in a group

Let $\langle G, * \rangle$ be a group and $a \in G$ with identity 'e' $\in G$. The least positive integer 'm' for which $a^m = e$ is called the order of the elt $a \in G$ and denoted by $O(a)$.

If no such ~~that~~ integer exists, then 'a' is of infinity order.

Example: $\langle G = \{1, -1, i, -i\}, \times \rangle$ – cyclic gp.

Then $O(1) = 1$, $O(i) = 4$, $O(-1) = 2$, $O(-i) = 4$

Theorem Let $\langle G, * \rangle$ be a finite cyclic group generated by an elt $a \in G$. If G is of order n, that is $|G| = n$, then $a^n = e$, so that $G = \{a, a^2, a^3, \ldots, a^n = e\}$. Furthermore, n is the least positive integer for which $a^n = e$.

Proof:- I step:

Let us assume that for some +ve integer $m < n$, $a^m = e$.

Since G is a cyclic group, any elt of G can be written as a^k, for some $k \in \mathbb{I}(\mathbb{Z})$

By Euclid's algorithm,

we can write $k = mq + r$, where 'q' is some integer and $0 \leq r < m$.

This means $a^k = a^{mq+r} = a^{mq} * a^r$

$$= (a^m)^q * a^r = e^q * a^r = e * a^r$$

$$a^k = a^r,$$

So that every elt can be expressed as a^r for some $0 \leq r < m$,

$\Rightarrow$ G has at least 'm' distinct elts

$\Rightarrow$ $|G| \leq m < n$

$\Rightarrow \Leftarrow$ which is a contradiction.

Hence, $a^m = e$ for $m < n$ is not possible.

Hence 'n' is the least +ve integer s.t $a^n = e$:

~~Now~~ Now to show that elts of G s.t $a, a^2, \ldots, a^n$ are all distinct, where $a^n = e$.

Suppose not, $\quad a^i = a^j$ for $i < j \leq n$ $\quad (a^{-j} = a^{-i}$ and $j - i < n)$

$$\Rightarrow a^j * a^{-i} = e$$

$$\Rightarrow a^{j-i} = e \Rightarrow j - i \geq n.$$

$$\Rightarrow\Leftarrow$$

which is a contradiction.

Subgroups and Homomorphisms

Subgroups Let $\langle G, * \rangle$ be a group and $S \subseteq G$ s.t it satisfies the following conditions

(i) $e \in S$ where e is the identity elt of $\langle G, * \rangle$.

(ii) for any $a \in S$, $a^{-1} \in S$.

(iii) For any $a, b \in S$, $a * b \in S$.

Then $\langle S, * \rangle$ is called subgroup of $\langle G, * \rangle$.

Examples:-

① Trivial subgroups.

$\langle \{e\}, * \rangle$ and $\langle G, * \rangle$ are trivial subgps of $\langle G, * \rangle$.

② Proper subgroups.

The subgroups other than trivial subgroups are called proper subgroups.

③ $\langle \overset{\text{even}}{E}, + \rangle$ is a subgroup of $\langle \mathbb{Z}, + \rangle$

$\langle \overset{\text{odd}}{O}, + \rangle$ is not a subgp of $\langle \mathbb{Z}, + \rangle$.

④ $\langle G' = \{1, -1\}, \times \rangle$ is a subgp of $\langle G = \{1, -1, i, -i\}, \times \rangle$.

Theorem A subset $S \neq \phi$ of G is a subgroup of $\langle G, *\rangle$ if and only if for any pair of elts $a, b \in S$, $a * b^{-1} \in S$.

Proof:-

Assume that S is a subgroup, it is clear that if $a, b \in S$, then $b^{-1} \in S$ and $a * b^{-1} \in S$.

To prove the converse, let us assume that $a, b \in S$ and $a * b^{-1} \in S$ for any $a, b \in S$.

To prove S is a subgroup.

Let $a, b \in S$.

ie, To prove $e \in S$, $a * b \in S$ and $a^{-1} \in S$.

Since, $a * b^{-1} \in S \Rightarrow a * b^{-1} = a * a^{-1} = e$

Take $b = a$ $\Rightarrow b^{-1} = a^{-1}$ $\Rightarrow a * b^{-1} = e \in S$

Now, $a \in S$ & $e \in S \Rightarrow e * a^{-1} \in S \Rightarrow a^{-1} \in S$.

Now $b \in S$, $e \in S \Rightarrow e * b^{-1} \in S \Rightarrow b^{-1} \in S$.

Now $a \in S$ and $b^{-1} \in S$

$\Rightarrow a * (b^{-1})^{-1} \in S \Rightarrow a * b \in S$

Hence $\langle S, *\rangle$ is a subgp of $\langle G, *\rangle$.

Group Homomorphism

Definition. Let $\langle G, *\rangle$ and $\langle H, \Delta\rangle$ be two groups. A mapping $g : G \rightarrow H$ is called a gp homomorphism from $\langle G, *\rangle$ to $\langle H, \Delta\rangle$, If for any $a, b \in G$,

(i) $g(a * b) = g(a) \Delta g(b)$

(ii) $g(e_G) = e_H$

(iii) $g(a^{-1}) = [g(a)]^{-1}$.

Definitions A group homomorphism 'g' is called a

(a) group monomorphism, If 'g' is 1-1.

(b) gp epimorphism, if 'g' is onto and

(c) gp isomorphism, if 'g' is both 1-1 & onto.

Isomorphic groups: $\cong$

If there is a gp isomorphism from a gp $\langle G, * \rangle$ to a group $\langle H, \Delta \rangle$, then $\langle G, * \rangle$ is isomorphic to $\langle H, \Delta \rangle$.

ie; $\langle G, * \rangle \cong \langle H, \Delta \rangle$.

Endomorphism

A group homomorphism from a group $\langle G, * \rangle$ to $\langle G, * \rangle$ is called an endomorphism.

Automorphism

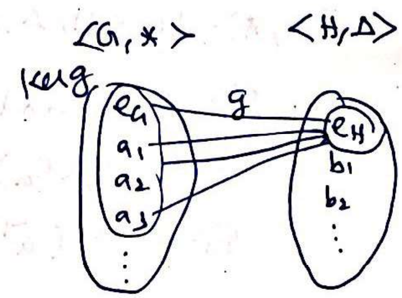
A group isomorphism from a gp $\langle G, * \rangle$ to a same group $\langle G, * \rangle$ is called an automorphism.

Kernel of a Homomorphism

Let 'g' be a gp homomorphism from $\langle G, * \rangle$ to $\langle H, \Delta \rangle$. The set of elements of G which are mapped into e_H (the identity of H) is called the kernel of the homomorphism 'g' and denoted by Ker(g).

i.e. ~~Kerg~~ $\ker(g) = \{ a \in G : g(a) = e_H \}$

Note: $\ker(g) \subseteq \langle G, * \rangle$.



Example Let $g : \langle \mathbb{Z}, + \rangle \to \langle \mathbb{Z}, + \rangle$ defined by

$$g(a) = a \quad \forall a \in \mathbb{Z}$$

↳ identity fn.

Then * 'g' is a gp homo.

* 'g' is a gp mono, gp epi, gp isomorphism, endo & automorphism.

Now, $\ker(g) = \{ a \in \mathbb{Z} : g(a) = 0 \} = \{0\}$

↳ $\langle \{0\}, + \rangle$ is a subgp of $\langle \mathbb{Z}, + \rangle$.

Theorem The Kernal of a homomorphism g from a group $\langle G, * \rangle$ to $\langle H, \Delta \rangle$ is a subgroup of $\langle G, * \rangle$.

Proof:- e_G, e_H are identity elts of G & H respectively.

(1) To prove: $e_G \in \langle \ker(g), * \rangle$

Since $g(e_G) = e_H \Rightarrow e_G \in \ker(g)$.

(2) To prove: For $a \in \ker g$, $a^{-1} \in \ker g$.

Since $a \in \ker g \Rightarrow g(a) = e_H$

Now. $g(a^{-1}) = [g(a)]^{-1} = (e_H)^{-1} = e_H$

$\Rightarrow g(a^{-1}) = e_H \Rightarrow a^{-1} \in \ker g$.

(3) To prove: For $a, b \in \ker g$, $a * b \in \ker g$.

Since, $a, b \in \ker g \Rightarrow g(a) = e_H$ & $g(b) = e_H$.

Now $g(a * b) = g(a) \Delta g(b) = e_H \Delta e_H = e_H$

$\Rightarrow a * b \in \ker g$

$\therefore$ $\ker g$ is a subgp of $\langle G, * \rangle$. R.[illegible]

Cosets and Lagrange's Theorem

Cosets Let $\langle H, * \rangle$ be a subgroup of $\langle G, * \rangle$. For any $a \in G$, the set aH defined by

$$aH = \{ a*h \mid h \in H \}$$ is called the left coset of H in G determined by the elt $a \in G$. The elt a is called the representative elt of the left coset aH.

Example

① Let $\langle G = \mathbb{Z}_4, +_4 \rangle$ be a gp s.t and $\langle H = \{[0],[2]\}, * \rangle$ is a subgp of $\langle \mathbb{Z}_4, +_4 \rangle$.

$+_4$	[0]	[1]	[2]	[3]
[0]	[0]	[1]	[2]	[3]
[1]	[1]	[2]	[3]	[0]
[2]	[2]	[3]	[0]	[1]
[3]	[3]	[0]	[1]	[2]

Then ~~the~~ distinct left cosets are

$\{[1],[3]\}$ and $\{[0],[2]\}$.

Now $aH = \{a*h \,;\, h \in H\}$ where $a \in G$.

If $a = [0]$, then

$$[0]H = \{[0]+_4[0], [0]+_4[2]\}$$
$$[0]H = \{[0],[2]\} = H.$$

If $a=[1]$, then $[1]H = \{[1]+_4[0], [1]+_4[2]\}$
$= \{[1],[3]\}$

If $a=[2]$, then $[2]H = \{[2]+_4[0], [2]+_4[2]\} = \{[2],[0]\}$
$= [0]H = H$.

If $a=[3]$, then $[3]H = \{[3]+_4[0], [3]+_4[2]\} = \{[3],[1]\}$
$= [1]H$.

② Ex. Find the left cosets. Consider $(\mathbb{Z}_{12}, \oplus)$. and $H = \{0, 4, 8\}$ is a subgroup.

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Lagrange's Theorem

(X) (X) The order of a subgroup of a finite group divides the order of the group.

Proof:- Let $\langle G, * \rangle$ be a finite group of order 'n'.

That is $|G| = n$,

and let $\langle H, * \rangle$ be a subgp of $\langle G, * \rangle$ and $|H| = m \leq n$.

To prove: 'm' divides 'n'. ie, n/m = an integer.

Property: Every left coset of H in G determined by any elt of G must have the same number of elt as the number of elts in H.

Hence, every left coset of H in G has exactly 'm' elements, and the left cosets partition the gp G. (Union of all the left cosets of H is G)

$\Rightarrow$ The number of left cosets of H in G must to n/m, and it is an integer.

$\therefore n = rm$

$\Rightarrow$ 'm' divides 'n'.

$\Rightarrow$ The order of a subgp of a finite gp divides the order of the group.

Corollary: ①

The order of any elt of a finite gp is a divisor of the order of the group.

Proof: Let $|G| = n$. Let $a \in G$, $a^m = e$ $o(a) = m$ order of a is the same as the order of the cyclic group $\langle a \rangle$. cyclic subgp is a subgp.

~~Proof~~:- ② $\langle G, * \rangle$ is a finite group of order 'n', then $a^n = e$ for any elt $a \in G$.

③ Every group of prime order is cyclic.

Let $|G| = p$ (prime)

Let $a \in G$ & $a \neq e$. $o(a)$ divides p.

$\therefore o(a)$ is 1 or p.

$\because a \neq e \therefore o(a) = p$

$G = \langle a \rangle$. Hence G is cyclic.

Normal subgroup:-

A Subgroup $\langle H, * \rangle$ of $\langle G, * \rangle$ is called a normal subgroup if for any $a \in G$, $aH = Ha$

(ie, the left & right coset of H in G generated by a are the same) or if $a^{-1} * h * a \in H$ for all $a \in G$, $h \in H$.

Example:-

Let $\langle \mathbb{Z}, + \rangle$ be the gp of integers and m be any +ve integer.

Then the set of multiples of 'm' forms a subgp which may denote by $\langle H_m, + \rangle$.

ie., $H_m = \{ mk \,;\, k \in N \}$

Since $\langle \mathbb{Z}, + \rangle$ is abelian, $\langle H_m, + \rangle$ is a normal subgp.

Theorem Let $\langle G, * \rangle$ and $\langle H, \Delta \rangle$ be groups and $g : G \to H$ be a homomorphism. Then the kernel of g is a normal subgp.

Proof:- WKT $\langle \ker g, * \rangle$ is a subgroup of $\langle G, * \rangle$

$$K = \ker g = \{ a \mid a \in G \text{ and } g(a) = e_H \}.$$

Let $a \in G$ and $k \in K \Rightarrow g(k) = e_H$.

To prove $a^{-1} * k * a \in K$ ie, $g(a^{-1} * k * a) = e_H$

For
$$\begin{aligned} g(a^{-1} * k * a) &= g(a^{-1}) \,\Delta\, g(k * a) \\ &= (g(a))^{-1} \,\Delta\, (g(k) \,\Delta\, g(a)) \\ &= (g(a))^{-1} \,\Delta\, (e_H \,\Delta\, g(a)) \\ &= (g(a))^{-1} \,\Delta\, g(a) \\ &= e_H \end{aligned}$$

$\therefore$ $\ker g$ is a normal subgp of $\langle G, * \rangle$.

Quotient Group (Factor group)

Let N be a normal subgroup of G. Then the group $\frac{G}{N}$ is called the Quotient gp of G modulo N.

Theorem (Fundamental theorem of gp homomorphism)

Let $f : \langle G, * \rangle \longrightarrow \langle H, \Delta \rangle$ be an epimorphism. Let K be the kernel of f. Then $\frac{G}{K} \cong H$.

Proof:- Define $\phi : \frac{G}{K} \longrightarrow H$ by $\phi(Ka) = f(a)$.

(i) ϕ is well defined.

Let $Kb = Ka$. Then $b \in Ka$

Hence $b = ka$ where $k \in K$

Now $f(b) = f(ka) = f(k) f(a) = e_H f(a) = f(a)$

$\therefore \phi(Kb) = f(b) = f(a) = \phi(Ka)$.

(ii) ϕ is 1-1.

For $\phi(Ka) = \phi(Kb) \Rightarrow f(a) = f(b)$

$\Rightarrow f(a) (f(b))^{-1} = e_H$

$\Rightarrow f(ab^{-1}) = e_H$

$\Rightarrow ab^{-1} \in K$

$\Rightarrow a \in Kb \Rightarrow Ka = Kb$.

(iii) ϕ is onto. Let $h \in H$. Since f is onto, $\exists\, a \in G$ s.t $f(a) = h$. $\therefore \phi(Ka) = f(a) = h$.

(iv) ϕ is homo

$\phi(KaKb) = \phi(Kab) = f(ab) = f(a) f(b)$

$= \phi(Ka)\phi(Kb)$.

$\therefore G/K \cong H$.

Similarly the set Ha defined by

$$Ha = \{h * a | h \in H\}$$

is called the *right coset* of H in G generated by $a \in G \cdot a$ is again called the representative (element) of Ha.

Lagrange's Theorem

The order of a subgroup of a finite group is a divisor of the order of the group.

Proof

Let aH and bH be two left cosets of the subgroup $\{H, *\}$ in the group $\{G, *\}$. Let the two cosets aH and bH be not disjoint.

Then let c be an element common to aH and bH i.e., $c \in aH \cap bH$.

Since, $c \in aH$, $c = a * h_1$, for some $h_1 \in H$ (1)

Since, $c \in bH$, $c = b * h_2$, for some $h_2 \in H$ (2)

From (1) and (2), we have

$$a * h_1 = b * h_2$$

$$\therefore \quad a = b * h_2 * h_1^{-1} \quad (3)$$

Let x be an element in aH

$$\therefore \quad x = a * h_3, \text{ for some } h_3 \in H$$

$$= b * h_2 * h_1^{-1} * h_3, \text{ using (3)}$$

Since H is a subgroup, $h_2 * h_1^{-1} * h_3 \in H$

Hence, (3) means $x \in b\,H$

Thus, any element in aH is also an element in bH. $\therefore$ $aH \subseteq bH$

Similarly, we can prove that $bH \subseteq aH$

Hence $aH = bH$

Thus, if aH and bH are not disjoint, they are identical.

$\therefore$ The two cosets aH and bH are disjoint or identical. (4)

Now every element $a \in G$ belongs to one and only one left coset of H in G, for,

$$a = ae \in aH, \text{ since } e \in H$$

i.e., $a \in aH$

$a \notin bH$, since, aH and bH are disjoint i.e., a belongs to one and only left coset of H in G i.e., aH. (5)

From (4) and (5), we see that the set of left cosets of H in G form a partition of G. Now let the order of H be m.

viz., let $H = \{h_1, h_2, \ldots, h_m\}$, where h_i's are distinct

Then $aH = \{ah_1, ah_2, \ldots ah_m\}$

The elements of aH are also distinct, for, $ah_i = ah_j \Rightarrow h_i = h_j$, which is not true.

Thus H and aH have the same number of elements, namely m.

In fact every coset of H in G has exactly m elements.

Now let the order of the group $(G, *)$ be n, i.e., there are n elements in G.

Let the number of distinct left cosets of H in G be p. [p is called the *index* of H in G.]

$\therefore$ the total number of elements of all the left cosets = pm = the total number of elements of G i.e. $n = p \cdot m$

i.e. m, the order of H is a divisor of n, the order of G.

Deductions

1. The order of any element of a finite group is a divisor of the order of the group.

   Proof

   Let $a \in G$ and let $O(a) = m$. Then $a^m = e$. Let H be the cyclic subgroup generated by a. Then $H = \{a, a^2, \ldots, a^m(= e)\}$. i.e., $O(H) = m$.

   By Lagrange's theorem,

   $O(H)$ is a divisor of $O(G)$

   $\therefore$ $O(a)$ is a divisor of $O(G)$

2. If G is a finite group of order n, then $a^n = e$ for any element $a \in G$.

   Proof

   If m is the order of a, then $a^m = e$. Then m is a divisor of n. i.e., $n = km$

   Now $a^n = a^{km} = (a^m)^k = e^k = e$.

3. Every group of prime order is cyclic.

   Proof

   Let $a(\neq e)$ be any element of G

   $\therefore$ $O(a)$ is a divisor of $O(G) = p$, a prime number

   $\therefore$ $O(a) = 1$ or p ($\because$ the divisors of p are 1 and p only)

   If $O(a) = 1$, then $a = e$, which is not true.

   Hence, $O(a) = p$. i.e., $a^p = e$

   $\therefore$ G can be generated by any element of G other than e and is of order p. i.e., the cyclic group generated by $a(\neq e)$ is the entire G.

   i.e., G is a cyclic group.

NORMAL SUBGROUP

Definition

A subgroup $\{H, *\}$ of the group $\{G, *\}$ is called a *normal subgroup*, if for any $a \in G$, $aH = Ha$ (i.e., the left and right cosets of H in G generated by a are the same)

Note $aH = Ha$ does not mean that $a * h = h * a$ for any $h \in H$, but it means that $a * h_1 = h_2 * a$, for some $h_1, h_2 \in H$.

Theorem

A subgroup $(H, *)$ of a group $(G, *)$ is a normal subgroup if and only if $a^{-1} * h * a \in H$ for every $a \in G$ and $h \in H$.

Proof

(i) Let $(H, *)$ be a normal subgroup of $(G, *)$.

Then $aH = Ha$ for any $a \in G$, by definition of normal subgroup.